Introduction to project management

A project is a temporary endeavour undertaken to create a unique product, service or result; project management applies knowledge, skills, tools and techniques to reach its objectives. This chapter sets the vocabulary used throughout the course and shows why IT projects fail and succeed.

PMBOK Guide 7th ed. **Process Groups Practice Guide 2022** **PRINCE2 7** **Case study: Realtime Quiz**

AGENDA

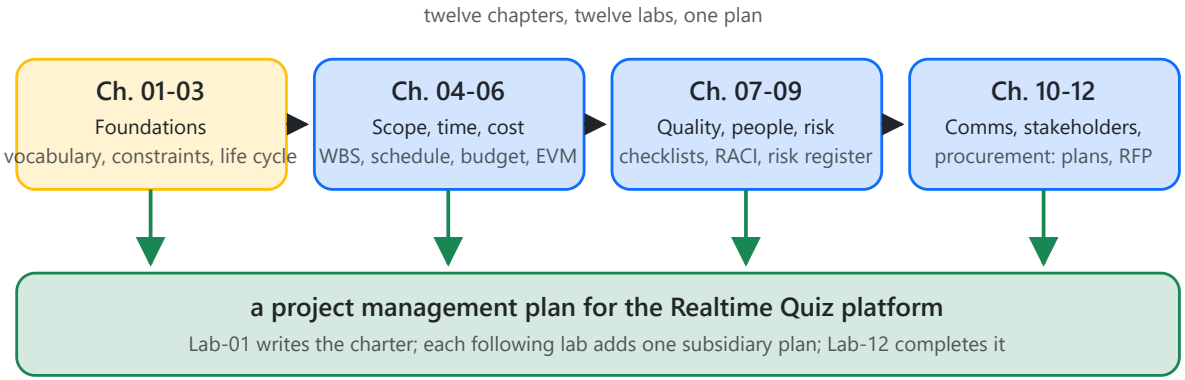
What we will cover

1. Projects, operations, programmes and portfolios
2. Characteristics of IT projects and why they are hard
3. The project manager role, competencies and ethics
4. Project success criteria and the history of IT project failure (Standish CHAOS, well-known failures)
5. Process groups: initiating, planning, executing, monitoring and controlling, closing
6. Knowledge areas and how the chapters of this course map to them
7. Organisational structures: functional, matrix, projectised, and the PMO
8. Project charter and business case
9. Project management software and information systems overview
10. Professional bodies and certifications (PMP, CAPM, PRINCE2, PSM)
11. Chapter Quiz
12. Lab-01

Click any item to jump directly to that topic.

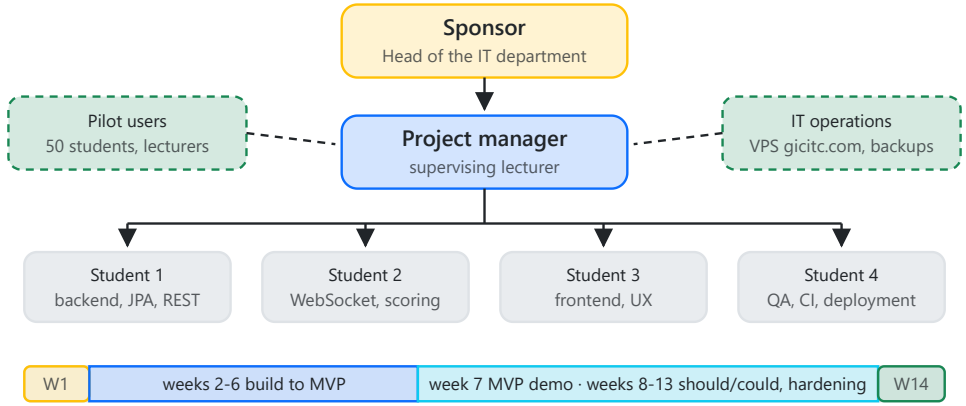
Why manage IT projects at all, and how this course is built

- In the Software Engineering course you **build** the Realtime Quiz platform. In this course you **manage** it: decide what gets built, by whom, by when, for how much, and prove it afterwards.
- Good code does not save a project that solves the wrong problem, runs late or exhausts its budget. Most IT project failures are management failures, not programming failures.
- Chapters 01 to 03 give the **vocabulary and the life cycle**; chapters 04 to 12 follow the **knowledge areas** in the order a project manager needs them.
- Every lab produces one part of a single, consistent **project management plan** for the Realtime Quiz project, in a Git repository.
- Today: what a project is, who the project manager is, why projects fail, and how a project is authorised through a charter and a business case.



The case study every lab manages: the Realtime Quiz platform

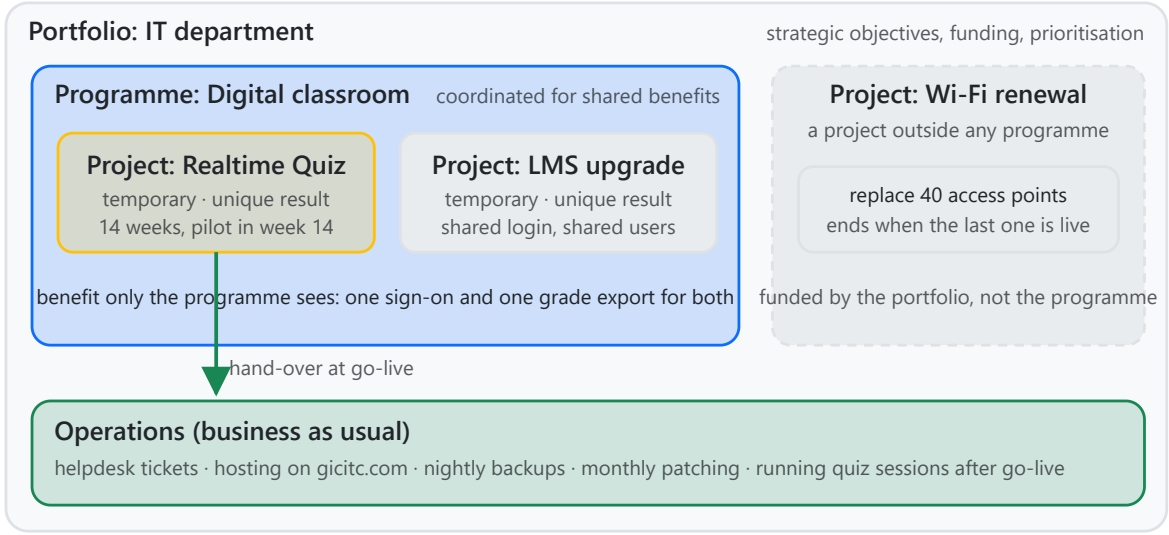
Fact	Value used in all twelve labs
Product	Web app for live classroom quizzes: instructors author quizzes and drive timed sessions, students join with a code, answer, see instant scores and a leaderboard (Spring Boot, WebSocket/STOMP, PostgreSQL)
Sponsor	Head of the IT department
Project manager	Supervising lecturer
Team	4 students (backend, real-time, frontend, QA/DevOps)
Duration	14-week semester, start 2026-09-29
Budget	Internal estimate in person-hours (490 h) plus about 200 USD cash: Linode VPS at gicitc.com (about 25 USD per month), a domain, optional paid tools
Go-live	Pilot with 50 students in week 14
MVP	All "Must" features (authoring, live session, timer, scoring, leaderboard, accounts and roles) demonstrated in week 7



14 weeks · charter in week 1 · pilot with 50 students and hand-over to operations in week 14

Projects, operations, programmes and portfolios PMBOK Guide 7th ed.

- **Project**: a temporary endeavour undertaken to create a unique product, service or result. Two tests: it has a *definite end*, and its output is *not a repeat* of last time.
- **Operations**: ongoing, repetitive work that keeps the organisation running (helpdesk, hosting, backups). Projects hand over their results to operations.
- **Programme**: related projects and programme activities managed together to obtain benefits that would not be available from managing them one by one.
- **Portfolio**: projects, programmes, subsidiary portfolios and operations grouped to achieve strategic objectives; managed by prioritising and funding.
- ISO 21502:2020 uses the same three levels; PRINCE2 7 calls the temporary organisation a *project* and its steady-state counterpart *business as usual*.



Worked example: project, operation or programme?

#	Initiative in the IT department	Class	Justification (temporary? unique? coordinated?)
1	Build the Realtime Quiz platform and pilot it with 50 students in week 14	Project	Definite end (week 14), result did not exist before
2	Run the nightly database backup of the gicitc.com VPS	Operation	Repeats every night, no end date, same output each time
3	Move all department servers to a new Linode region before the December maintenance window	Project	Deadline, one-off migration, ends when the last server is moved
4	Answer helpdesk tickets within the 4-hour service level	Operation	Continuous service; measured by a level, not by a finish
5	"Digital classroom": quiz platform, LMS upgrade and lecture-capture rollout with one login and one grade export	Programme	Several related projects; the shared login is a benefit no single project delivers
6	Apply the monthly security patches to the VPS	Operation	Repetitive, scheduled, part of keeping the service running
7	Add multi-node scaling to Realtime Quiz next year (RabbitMQ relay, Redis registry)	Project	A new, unique result with its own start and end; a follow-up project, not maintenance
8	Support the quiz platform after go-live: reset passwords, restart, monitor	Operation	Ongoing; it is what the project hands over to

💡 **Two-question test.** Does it end? Is the result new? Yes to both: project. No to either: operation. Several related projects that only make sense together: programme.

⚠ **Grey zone.** A "release 1.1" with bug fixes only is maintenance (operation). A release that adds a new capability with a deadline and a budget is a small project. Ask who authorised it and what "done" means.

✓ **Lab-01 Task 1** gives you ten initiatives to classify with this justification column.

TOPIC 2

Characteristics of IT projects and why they are hard

- The product is **intangible**: you cannot see 60 % of a login feature the way you see 60 % of a wall. Progress is reported, not observed.
- **Requirements move**: users discover what they want when they see a first version; technology and vendors change under the project.
- **Estimation is weak**: every feature hides integration, security, error handling and tests that the one-line description omits.
- **Specialists**: backend, real-time, frontend, database, security and operations rarely sit in one head; communication load grows with team size.
- Brooks (No Silver Bullet, 1987) named four *essential* difficulties of software: complexity, conformity, changeability, invisibility. They explain most IT project pain.

Brooks' four essential difficulties, seen in the Realtime Quiz project

Complexity	Conformity	Changeability	Invisibility
no two parts alike; grows faster than size	must fit rules made by others	software is asked to change because it can	no geometry to look at; progress is claimed
50 students answer in a 20 s window: races, locks, late answers	campus proxy blocks WebSocket; student data rules; CSP	"could we add chat, a question bank, an admin screen?"	"timer is 90 % done" for three weeks
PM: decompose (WBS)	PM: constraints, risks	PM: change control	PM: demos, EVM, tests

Source: F. Brooks, "No Silver Bullet" (1987); management responses are this course's chapters 04, 06, 09

TOPIC 2

Worked example: one line of feature list, thirty hours of work

```
Requirement as written: "Timer functionality for each question"  (1 line)

What it hides once decomposed                                 estimate
  server-side deadline per question                              4 h
  client clock-offset measurement on connect                     3 h
  250 ms grace period and rejection of late answers              3 h
  reconnect: resume the running countdown                        8 h
  concurrency test: 100 answers arriving at the deadline         6 h
  host console and play page countdown display                   6 h
                                                                ------
Total                                                           30 h

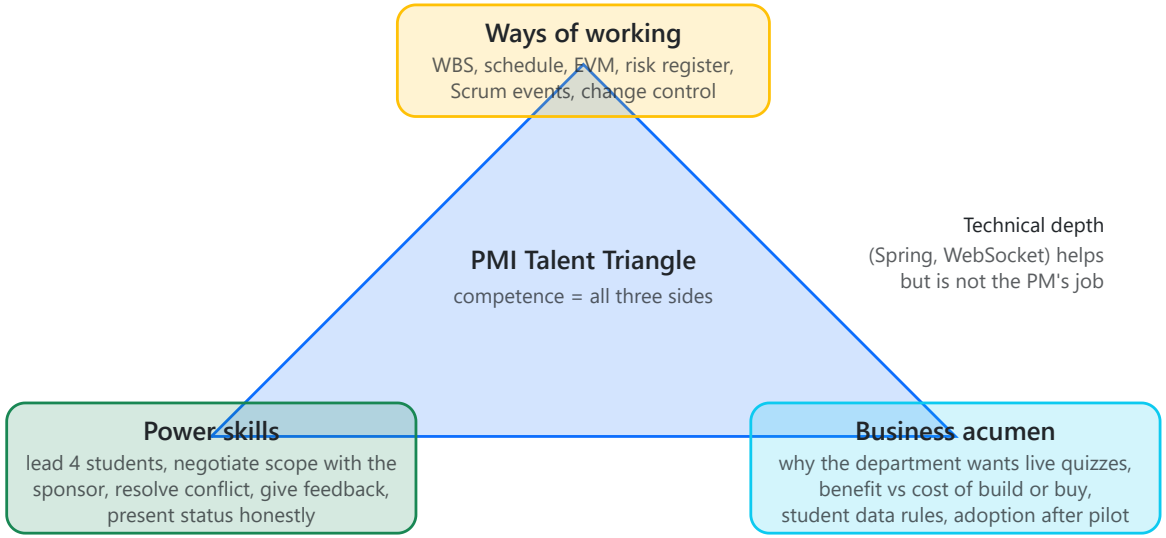
One line of feature list became six tasks and 30 hours. Estimates
made from feature lists are optimistic because the work is invisible
until someone decomposes it (Chapter 04, WBS; Chapter 05, PERT).
```

Characteristic	In Realtime Quiz	What the PM does
Intangible product	"Scoring works" cannot be seen	Weekly demo of running software
Moving requirements	"Could" list grows every review	Freeze MVP scope at week 4, change log
Weak estimates	Timer: 1 line, 30 h	Decompose before estimating, keep a reserve
Specialists	4 students, 4 skill sets	RACI, pairing, shared definition of done
Dependence on others	VPS, campus network, JDK and Spring versions	Assumptions log, risk register

⚠ **"90 % done" syndrome.** Invisible work makes the last 10 % take as long as the first 90 %. Accept only demonstrated, tested increments as progress.

The project manager: role and competencies PMI Talent Triangle

- The **project manager** is assigned by the sponsor to lead the team that is responsible for achieving the project objectives (PMBOK Guide 7th ed.).
- The PM **integrates**: turns sponsor intent into a plan, keeps scope, schedule, cost, quality, people, risk and communication consistent, and reports the truth.
- The PM is *not* necessarily the best programmer. In the Realtime Quiz project the lecturer manages; the students own the technical decisions inside the plan.
- PMI's **Talent Triangle**: ways of working (technical project management, predictive and agile), power skills (leadership, communication, negotiation), business acumen (why the project exists, its benefits and risks).
- Time split of a typical PM: most of the day is communication; the plan is updated in the gaps.



Ethics: four values and a worked dilemma

PMI Code of Ethics and Professional Conduct

Value	Means	In a student project
Responsibility	Own your decisions and their consequences	The PM signs the estimate and owns the recovery plan
Respect	Regard for people, resources, environment	Listen to the QA student's warning before the demo
Fairness	Impartial, objective decisions; no conflicts of interest	Credit and grade contributions as they were made
Honesty	Understand and report the truth	Report the real coverage number, not the target

The code distinguishes *aspirational* standards (what we strive for) from *mandatory* standards (what is required, such as not misrepresenting facts). Most exam questions and real dilemmas are about honesty under pressure.

```
Situation (week 7): the sponsor wants the status report to say "MVP on
track" for the faculty meeting. Reconnect-and-resume still fails and the
concurrency test is red.

Options
  1. Report "on track" as asked; fix it quietly next week.
  2. Report "amber: MVP demo at risk, 2 defects open, recovery plan attached".
  3. Report "red" and ask to move the demo.

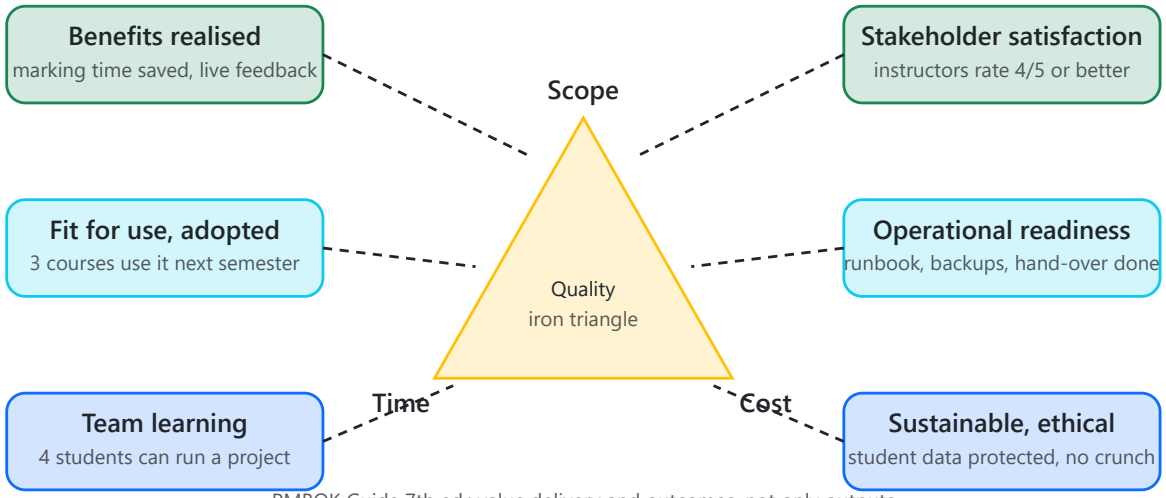
Values check
  Honesty         option 1 misstates the status              -> excluded
  Responsibility  own the recovery plan, not just the news   -> 2 or 3
  Fairness        the team is not blamed for a plan the PM accepted
  Respect         raise it with the sponsor first, privately, then in writing

Decision: option 2. The report names the two defects, their owner, the
retest date (week 8) and the trigger for moving the demo (test still red
on Thursday). The sponsor keeps the choice; the PM keeps the facts.
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TOPIC 4

What counts as success? Beyond the iron triangle

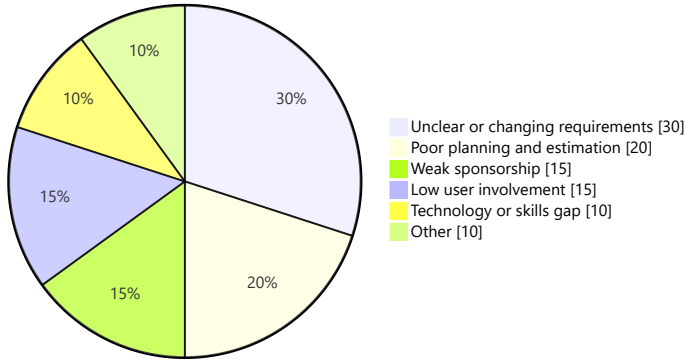
- The classic test: delivered **on scope, on time, on budget** (the iron triangle, Chapter 02). Necessary, but not sufficient: a product nobody uses is a failure that met its plan.
- PMBOK Guide 7th ed. asks for **outcomes and value**: did the intended benefits happen, are stakeholders satisfied, is the result fit for use?
- Success criteria must be written **before** the work starts, be **measurable**, and be agreed with the sponsor. They go into the charter.
- Realtime Quiz criteria: pilot in week 14 with 50 students and 0 blocking defects; broadcast latency at most 500 ms (95th percentile); coverage at least 80 %; instructor satisfaction at least 4 of 5; within 490 h and 200 USD.
- Different stakeholders weight the criteria differently: students want a smooth session, the department wants adoption, the lecturer wants learning.



PMBOK Guide 7th ed.: value delivery and outcomes, not only outputs

The history of IT project failure: CHAOS reports and famous cases

Why IT projects fail (illustrative, CHAOS-style categories)



The Standish Group has published **CHAOS** reports since 1994. Across editions roughly a third of projects are rated successful (on time, on budget, satisfactory result), about half challenged, the rest failed or cancelled; small projects succeed far more often than large ones. The percentages above are illustrative rounding of the recurring failure categories, not a quotation of one edition.

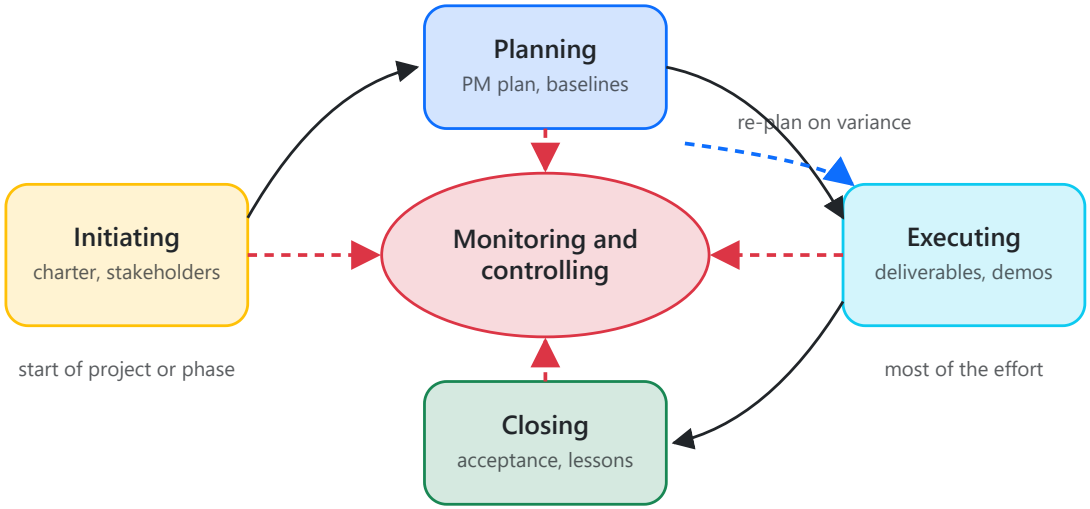
Case (widely reported)	Year	What happened	Management lesson
Denver airport baggage system	1995	Automated baggage system delayed the airport opening by about 16 months	Complexity and scope underestimated; schedule set before design
Hershey ERP go-live	1999	Cut-over in the peak season; orders for Halloween not shipped	Compressed schedule, big-bang cut-over, no fallback
FBI Virtual Case File	2005	Abandoned after roughly 170 million USD	Changing requirements, weak oversight, no working increments
UK NHS National Programme for IT	2011	Dismantled; cost estimates around 10 billion GBP	Too large, top-down, low clinician involvement
Knight Capital deployment	2012	A faulty release lost about 440 million USD in 45 minutes	Release and operations controls are project deliverables too
Healthcare.gov launch	2013	Site failed under load on launch day	No end-to-end testing, dozens of contractors, no single owner

⚠ Every row is a management failure first: requirements, sponsorship, testing, cut-over planning. The Lab-01 challenge asks you to analyse one such case with this chapter's concepts.

Five process groups

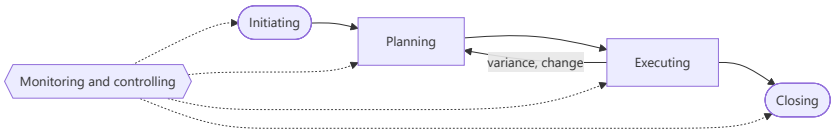
Process Groups Practice Guide 2022

- **Initiating**: define and authorise the project or a phase; outputs: charter, stakeholder register.
- **Planning**: establish scope, refine objectives, define the course of action; output: the project management plan and its subsidiary plans.
- **Executing**: do the work in the plan; manage people, quality, communication, procurement.
- **Monitoring and controlling**: track, review, regulate progress; compare against baselines; manage changes. Runs from the first day to the last.
- **Closing**: formally complete the project or phase: acceptance, lessons learned, hand-over, release of the team.
- They are **not phases**: they interact and repeat inside each phase of the life cycle (Chapter 03). The 2022 guide groups 49 processes this way.



dashed red: monitoring touches every group · groups repeat in every phase

Worked example: the five groups mapped to the 14-week semester



💡 Mermaid `flowchart` source is teaching material: Lab-01 Task 5 asks for a Mermaid `gantt` of this mapping.

Process group	Weeks	Realtime Quiz outputs (built in this course's labs)
Initiating	1	charter, stakeholder list, business case (Lab-01)
Planning	1-3	scope statement and WBS, schedule, budget, quality, RACI, risk, communication, stakeholder and procurement plans (Labs 02-12 build them in class)
Executing	3-13	Realtime Quiz increments; weekly demo; tagged commit
Monitoring and controlling	1-14	weekly status: schedule and cost variance, risk review, change log, quality checks
Closing	14	pilot with 50 students, acceptance by the sponsor, lessons learned, hand-over to IT operations

Rule of thumb: planning does not end in week 3. Every variance found by monitoring re-opens planning; every accepted change re-baselines the plan.

Ten knowledge areas, twelve chapters, one plan

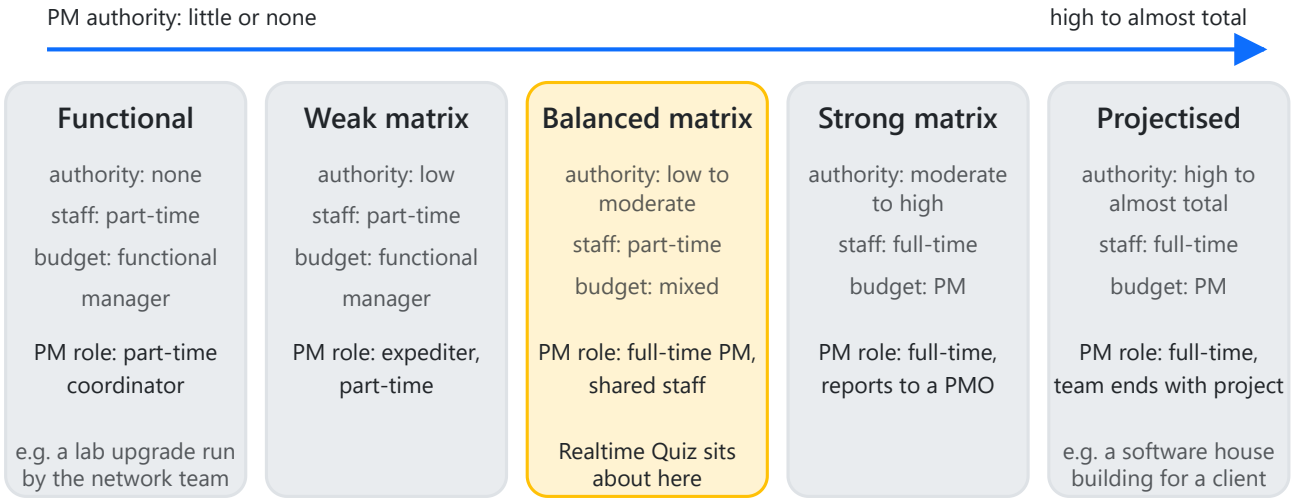
Knowledge area	Question it answers	Chapter	Lab deliverable in the plan
Integration	How do the parts fit together? Charter, PM plan, change control, closing	01, 03	Charter (Lab-01); life cycle selection and tailoring (Lab-03)
Scope	What is, and is not, included?	02, 04	Constraint analysis (Lab-02); scope statement, WBS, WBS dictionary (Lab-04)
Schedule	When?	02, 05	Activity list, network diagram, critical path, Gantt (Lab-05)
Cost	How much?	02, 06	Estimates, budget, earned value baseline (Lab-06)
Quality	How good, and how do we know?	07	Quality management plan, checklists (Lab-07)
Resource	Who, and with what?	08	Organisation chart, RACI, staffing plan (Lab-08)
Risk	What could go wrong or right?	09	Risk register and response plans (Lab-09)
Communications	Who needs to know what, when, how?	10	Communication management plan (Lab-10)
Stakeholder	Who cares, and how do we engage them?	11	Stakeholder register and engagement plan (Lab-11)
Procurement	What do we buy, from whom, under which contract?	12	Procurement plan, RFP, contract evaluation (Lab-12)

- A **knowledge area** is a field of specialisation with its own processes, inputs, tools and outputs; the ten areas come from the PMBOK Guide 6th ed. and are kept in the Process Groups Practice Guide (2022).
- The PMBOK Guide 7th ed. reorganised the same content into 12 principles and 8 **performance domains** (stakeholders, team, development approach and life cycle, planning, project work, delivery, measurement, uncertainty). This course teaches by knowledge area because a plan is written that way.
- Chapter 02 (triple constraint) cuts across scope, schedule and cost; Chapter 03 chooses the life cycle that shapes all the others.

☑ Each lab from Lab-01 on produces one of the documents in this table, filled in over the semester.

Organisational structures: how much authority does the PM have?

- **Functional**: staff report to a department head; the "PM" is a coordinator with little or no authority and part-time staff.
- **Matrix**: staff report to both a functional manager and a PM. Weak, balanced or strong depending on who controls priorities and budget.
- **Projectised**: the team reports only to the PM, who controls the budget; staff are re-assigned when the project ends.
- The structure is a **constraint** the PM inherits: it decides how work is requested, who resolves conflicts and how fast decisions happen.
- Realtime Quiz: students belong to a course (functional home), the lecturer-PM controls their project hours and the 200 USD cash: a **balanced to strong matrix**.



in your own words after PMBOK Guide 6th ed.; PMBOK 7 adds organic, virtual and hybrid forms
the further right, the faster decisions, the higher the cost of idle specialists

The project management office (PMO): three types

PMO type	Degree of control	What it does	Fits when
Supportive	Low	Consultative: templates, training, lessons-learned repository, access to information from other projects	Many small projects, experienced teams, low risk
Controlling	Moderate	Requires compliance: a mandated framework, templates, governance gates, reporting format	Regulated or audited environments, mixed maturity
Directive	High	Manages the projects directly; project managers report to the PMO	Large, strategic, cross-department programmes

- A PMO standardises governance and shares resources, methods, tools and techniques across projects.
- It is the natural home of the **portfolio** view: which projects run, which wait, which stop.
- Danger sign: a PMO that produces reports nobody reads. Its value is measured by projects finished, not templates issued.

```
Worked example: a supportive PMO for the IT department's capstones

Question  Which PMO type for 6-10 student capstone projects a year,
          each 14 weeks, each with a lecturer as PM?
Answer    Supportive. Low risk, short, experienced supervisors;
          control gates would cost more than they save.

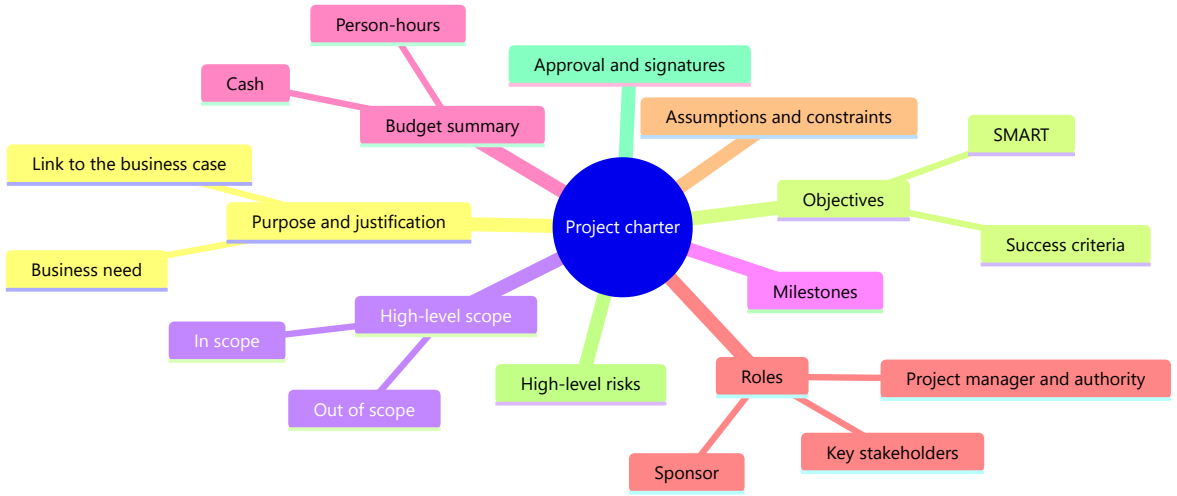
What the PMO gives the Realtime Quiz team
  templates       charter, WBS dictionary, risk register, status report
                  (this course's lab templates, in a shared repository)
  standards       Markdown + Git for every plan; one status format
  knowledge       lessons learned from last year's capstones
  services        a Linode VPS per project, a domain, a backup job
  reporting       one page per project, monthly, to the Head of IT

What it does NOT do
  approve scope changes (sponsor), grade students (lecturer),
  assign work inside the team (PM and team)
```

TOPIC 8

The project charter and the business case PMBOK Guide · PRINCE2 7

- The **business case** answers "why should we spend money on this?": problem, options, costs, benefits, risks, recommendation. PRINCE2 7 checks it at every stage boundary: *continued business justification*.
- The **project charter** is issued by the **sponsor** (someone outside the project with the authority to fund it). It formally authorises the project, names the project manager and states the PM's authority.
- It is short (1-3 pages) and high-level: purpose, measurable objectives, success criteria, high-level scope and requirements, milestones, budget summary, key stakeholders, assumptions, constraints, risks, approval.
- Anything not in the charter is negotiated later in planning; anything in it is a commitment.
- Business case first, charter second, project management plan third (Labs 01, then 02-12).



Worked example: Realtime Quiz charter, part 1 (purpose, objectives, scope)

```
# Project charter: Realtime Quiz platform           v1.0 · 2026-09-29

## Purpose and justification
Lecturers lose 1-2 hours a week marking paper quizzes and get no live
picture of class understanding. The platform runs live, timed quizzes
in class with instant scoring, and gives the department a reusable tool
(see business-case.md, option C).

## Objectives (SMART) and success criteria
| #  | Objective                                | Success criterion    |
|----|------------------------------------------|----------------------|
| O1 | MVP (all Must features) demonstrated     | demo passes, week 7  |
| O2 | Pilot with 50 students in one live session | 0 blocking defects   |
| O3 | Broadcast latency during the pilot       | <= 500 ms, 95th pct  |
| O4 | Automated test coverage                  | >= 80 % line, JaCoCo |
| O5 | Instructor satisfaction after the pilot  | >= 4 / 5 in survey   |

## High-level scope
In:  quiz authoring, live sessions with timer, scoring, leaderboard,
     accounts and roles, CSV/PDF export, responsive web UI, VPS deploy
Out: native mobile apps, LMS integration, free-text questions,
     multi-node scaling (documented as a future path only)
```

SMART	Check on O2 "Pilot with 50 students"
Specific	one live session, real students, real quiz
Measurable	50 participants, 0 blocking defects
Achievable	capacity NFR is 100 per session; 50 is half
Relevant	proves the purpose: a live quiz in a real class
Time-bound	week 14 of the semester

⚠ **Not an objective:** "Build a great quiz app" (no measure, no date). **Not scope:** "Use Spring Boot" (that is a constraint or a design decision).

💡 Objectives say *what* will be true at the end; success criteria say *how we will know*. Every criterion needs a number and a date.

Worked example: Realtime Quiz charter, part 2 (milestones, budget, roles, authority)

```
## Milestones (14-week semester, start 2026-09-29)
| Week | Milestone                                    | Process group |
|------|----------------------------------------------|---------------|
| 1    | Charter approved, team formed                | Initiating    |
| 3    | Project management plan baselined            | Planning      |
| 4    | Accounts, security and quiz authoring done   | Executing     |
| 7    | MVP demo: live session, timer, leaderboard   | Executing     |
| 10   | Analytics, export, notifications done        | Executing     |
| 12   | Hardening: 100-student load test passed      | Executing     |
| 14   | Pilot with 50 students, lessons learned      | Closing       |

## Budget summary
Effort: 4 students x 14 weeks x 8 h = 448 h; PM 14 x 3 h = 42 h; 490 h
Cash:   hosting 4 x 25 USD = 100; domain 15; paid tools 60; reserve 25
        total 200 USD (Linode VPS at gicitc.com)

## Roles and key stakeholders
Sponsor: Head of the IT department     PM: supervising lecturer
Team: 4 students (backend, real-time, frontend, QA/DevOps)
Key stakeholders: 50 pilot students, course lecturers, IT operations,
                  Software Engineering lecturer, data protection officer

## Assumptions, constraints, authority
Assumes:     students give 8 h/week; the VPS is available from week 2
Constraints: 14 weeks fixed; 200 USD cash cap; student data stays on-site
Authority:   the PM may re-plan within the budget; scope changes and
             extra cash need the sponsor's written approval
Approved: _____________________ (sponsor)      Date: ____________
```

Budget line	Calculation	Result
Student effort	4 × 14 weeks × 8 h	448 h
PM effort	14 weeks × 3 h	42 h
Total effort	448 + 42	**490 h**
Hosting	4 months × 25 USD	100 USD
Domain	one year	15 USD
Paid tools	optional (icons, monitoring)	60 USD
Reserve	about 14 % of 175	25 USD
Total cash	100 + 15 + 60 + 25	**200 USD**

The charter budget is a **summary** with an order of magnitude. Lab-06 replaces it with a bottom-up estimate; if the two disagree by more than the reserve, the sponsor is told.

Worked example: the business case with a benefit / cost comparison

```
Options (figures are course assumptions, not vendor quotes)
A. Do nothing: paper quizzes, manual marking            cost 0, benefit 0
B. Buy a quiz SaaS: 10 lecturers x 15 USD x 12 months   = 1,800 USD / year
C. Build Realtime Quiz: 490 h x 10 USD (notional rate)  = 4,900 USD one-off
                        + hosting 12 x 25 USD           =   300 USD / year

Annual benefit of B or C
  marking time saved: 10 lecturers x 1 h x 30 weeks = 300 h x 10 = 3,000 USD
Extra benefit of C over B
  SaaS fee avoided                                              1,800 USD
  own data, no per-seat limit, capstone learning outcome      not priced

Payback of C = 4,900 / (3,000 + 1,800 - 300) = 4,900 / 4,500 = 1.1 years
Recommendation: C, with B as the fallback if the MVP demo fails in week 7
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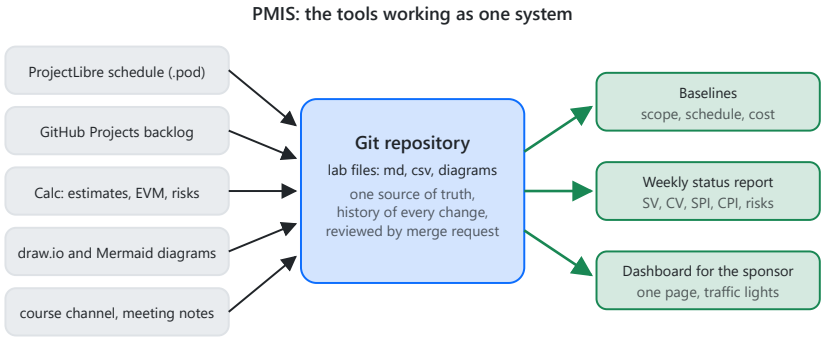
Business case section	Realtime Quiz content
Problem	Marking load, no live feedback, no reusable tool
Options	A do nothing, B buy SaaS, C build
Costs	One-off effort, recurring hosting or fees
Benefits	Time saved, fees avoided, data ownership, learning
Risks	MVP slips (C); vendor lock-in and data location (B)
Recommendation	C, reviewed at the week-7 gate

⚠ Student hours are not paid, so the 10 USD notional rate makes options comparable; say so in the document. A business case with unpriced benefits only is an opinion.

Project management software and the PMIS

Category	Typical tools	Used for	In this course
Scheduling	Microsoft Project 2024, ProjectLibre 1.9	WBS, Gantt, critical path, resource levelling	ProjectLibre (Labs 05, 06)
Agile tracking	Jira Cloud, GitHub Projects, Trello	Backlog, sprints, burndown, kanban	GitHub Projects
Spreadsheets	Excel 365, LibreOffice Calc 24.x	Estimates, budgets, earned value, risk register	Calc or Excel, CSV in Git
Diagramming	draw.io 24.x, Mermaid, PlantUML	Org charts, stakeholder maps, network diagrams	draw.io, Mermaid
Documents and versions	Git and GitHub, Markdown, Confluence	Plans, change history, review	Git + Markdown, one repo per student
Communication	Teams, Slack, e-mail, Zoom	Status, decisions, escalation	Course channel
Enterprise PPM	Planview, Broadcom Clarity, Jira Align	Portfolio, resource pools, financials	Recognition only

Prefer the free tools so every student can complete the labs on their own laptop; the commercial products are shown so you recognise them in a job advertisement.

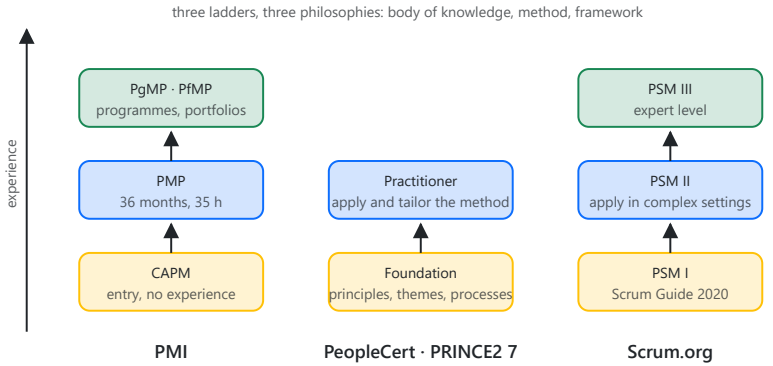


a PMIS is the integrated set, not one product; the plan lives where the code lives

Professional bodies and certifications

Body	Certification	Audience	Path (verify current rules on the body's website)
PMI (Project Management Institute)	**CAPM**	Students, junior team members	Secondary degree plus 23 h of PM education; one exam; no experience needed
PMI	**PMP**	Experienced project managers	36 months leading projects (with a degree) plus 35 h of education; exam based on the Exam Content Outline; renewal with 60 PDUs every 3 years
PMI	PMI-ACP, PgMP, PfMP	Agile practitioners, programme and portfolio managers	Experience in the discipline plus exam
PeopleCert (Axelos)	**PRINCE2 7** Foundation, Practitioner	Method-based roles, common in the UK, Europe and public sector	Foundation exam; Practitioner requires Foundation
Scrum.org	**PSM** I, II, III	Scrum masters, team members	Online exam on the Scrum Guide 2020; no course required
Scrum Alliance	CSM, CSPO	Scrum masters, product owners	Two-day course plus exam; renewal
IPMA	Level D to A	Competence-based, national associations	Assessment against the ICB4 competence baseline
CompTIA	Project+	IT professionals who also run projects	One exam

Why care as a student: CAPM and PSM I are reachable during the degree; vocabulary in this course (charter, WBS, EVM, process groups) is the exam vocabulary.



WRAP-UP

Best practices and common mistakes when starting a project

1 Get a named sponsor before you start

No charter, no project. A project without a sponsor who funds it and settles disputes drifts until it is cancelled. The sponsor signs the charter; the PM does not sign their own authority into existence.

2 Write success criteria with numbers and dates

"Pilot with 50 students in week 14, 0 blocking defects" can be checked. "A successful launch" cannot. Every criterion in the charter must be measurable by someone other than the team.

3 State what is out of scope

The exclusions list (no mobile apps, no LMS integration) prevents more arguments than the inclusions list. Changeability is essential to software; the charter is your first defence.

✕ Mistake: confusing the charter with the plan

The charter is 1-3 pages of intent and authority. Task lists, Gantt charts and RACI belong in the project management plan (Labs 02-12). A 40-page charter is a plan nobody approved.

✕ Mistake: treating process groups as phases

"We finished planning in week 3" is a warning sign. Monitoring re-opens planning every week; each phase has its own initiating and closing. Phases are the life cycle (Chapter 03).

✕ Mistake: reporting the plan instead of the facts

Invisibility tempts everyone to say "90 % done". Report demonstrated increments and test results; keep the honesty value of the PMI code even when the sponsor prefers green.

Chapter quiz 10 questions

1. Which of these initiatives in the IT department is a project?

- Running the nightly database backup of the VPS
- Answering helpdesk tickets within 4 hours
- Building the Realtime Quiz platform for a pilot in week 14
- Applying the monthly security patches

2. The "Digital classroom" initiative groups the quiz platform, an LMS upgrade and a lecture-capture rollout so they share one login. What is it?

- A portfolio
- A programme
- A large project
- An operation

3. In which organisational structure does the project manager have the highest authority and control the budget?

- Functional
- Weak matrix
- Balanced matrix
- Projectised

4. Which process group runs from the first week of the project to the last?

- Initiating
- Planning
- Monitoring and controlling
- Closing

5. Who issues the project charter?

- The project manager
- The sponsor, outside the project, with authority to fund it
- The development team
- The PMO templates committee

6. Four students give 8 hours a week for 14 weeks. How many person-hours of student effort does the charter budget contain?

- 112 h
- 448 h
- 490 h
- 560 h

7. Business case: build costs 4,900 USD one-off plus 300 USD per year; it saves 3,000 USD of marking time and avoids 1,800 USD of SaaS fees per year. Payback period?

- About 0.6 years
- About 1.1 years
- About 1.6 years
- About 2.7 years

8. Hosting costs 25 USD a month for 4 months, the domain 15 USD and optional tools 60 USD. Cash cost before any reserve?

- 100 USD
- 160 USD
- 175 USD
- 200 USD

9. Which of these is NOT one of the four values of the PMI Code of Ethics and Professional Conduct?

- Responsibility
- Respect
- Efficiency
- Honesty

10. A PMO that manages projects directly and to which project managers report is called:

- Supportive
- Controlling
- Directive
- Consultative

WRAP-UP

Summary

A project is temporary and produces a unique result; operations are ongoing and repetitive; programmes coordinate related projects for shared benefits; portfolios align everything with strategy.

IT projects are hard because software is complex, must conform to others' rules, is asked to change, and is invisible until demonstrated (Brooks).

The project manager integrates and communicates; competence spans ways of working, power skills and business acumen, under the PMI values of responsibility, respect, fairness and honesty.

Success is more than scope, time and cost: benefits realised, satisfied stakeholders, a product that is used and can be operated.

The CHAOS reports and the famous failures point at requirements, sponsorship, testing and cut-over: management causes, not coding causes.

Five process groups interact and repeat in every phase; monitoring and controlling runs from the first week to the last.

Ten knowledge areas structure the project management plan; chapters 02-12 fill one section each.

Organisational structure sets the PM's authority from functional to projectised; a PMO can support, control or direct.

The business case justifies the investment; the charter, issued by the sponsor, authorises the project, names the PM and fixes measurable success criteria.

Open Lab-01: 5 tasks + 1 challenge

Next chapter: 02 · The triple constraint (scope, time, cost)